

Data Analytics (DA)

Module -5

Python DA Assignment 1—Numpy and Pandas

Assignment Task Overview

This assignment focuses on performing **data analysis using NumPy and Pandas**.

The objective is to:

- Work with **NumPy arrays** for numerical computations
- Use **Pandas Series and DataFrame** for data manipulation and analysis
- Apply **indexing, slicing, filtering, and aggregation techniques**
- Understand real-world data handling through temperature and transaction datasets

You will analyze temperature data using NumPy and perform structured data analysis using Pandas.

Tasks:

NumPy Array Operations:

Scenario: You are given a list of daily average temperatures (in degrees Celsius) recorded over two weeks. Your task is to use NumPy to analyze this data.

1. Create a 1D NumPy Array:

Create a 1D NumPy array named `temperatures_w1` for Week 1 with the following values:

`[22.5, 25.3, 20.8, 23.4, 26.1, 24.8, 21.9]`

2. Inspection and Properties:

- Inspect the shape, data type, and the number of elements in the array. Inspect the shape of the `temperatures_w1` array.

3. Array Operations:

- Convert the `temperatures_w1` from Celsius to Fahrenheit using the formula:
`Fahrenheit = (Celsius * 9/5) + 32`
- Find the maximum, minimum, and mean temperatures for the week.

4. Array Slicing and Indexing:

- Extract the temperatures for the first three days of the week.
- Extract the temperatures for the weekend (last two days).
- Extract the temperatures for the middle three days of the week.

5. Create a 2D Array:

Create a 2D NumPy array named `temperatures` where each row represents temperatures for a different week:

★ Row 1: `[22.5, 25.3, 20.8, 23.4, 26.1, 24.8, 21.9]` (Week 1)

★ Row 2: `[19.2, 22.5, 21.3, 24.0, 23.5, 22.8, 20.1]` (Week 2)

6. Inspect and Slice the 2D Array:

- Inspect and print the shape, data type, and total number of elements in the `temperatures` array.
- Extract the temperatures for each week and the weekends (last two days) for both weeks.

Pandas Series:

1. Creating Pandas Series:

- Create a Pandas Series named `marks` with the following data:
 - Marks: `95, 92, 89, 85, 80`
 - Custom Indices: `'Rank1', 'Rank2', 'Rank3', 'Rank4', 'Rank5'`

2. Indexing and Slicing:

- Use the integer index position to access the mark of the 1st rank student.
- Use the `loc` accessor to retrieve the marks of the top 3 ranks by specifying their index labels.
- Use the `iloc` accessor to retrieve the mark of the 3rd-ranked student.
- Apply a boolean mask to filter the Series and retrieve the ranks where the marks are greater than 90.

3. Manipulating Series:

- Modify the mark of the student with the 1st rank to 100.
- Remove the entry corresponding to the last rank from the Series.
- Compute the CGPA by dividing each mark in the Series by 10.

Pandas DataFrame:

1. Creating Pandas DataFrame:

- Create a Pandas DataFrame named `transactions` with the following data: ➤
 - TransactionID:** 101, 102, 103, 104, 105, 106, 107, 108, 109, 110
 - **ProductCategory:** 'Electronics', 'Clothing', 'Electronics', 'Furniture', 'Clothing', 'Electronics', 'Furniture', 'Clothing', 'Furniture', 'Electronics'
 - **Region:** 'North', 'South', 'North', 'East', 'West', 'North', 'East', 'West', 'South', 'North'
 - **Amount:** 200, 150, 300, 450, 200, 250, 300, 180, 350, 400

2. Data Exploration:

- Display the `transactions` DataFrame and its basic information, including `head`, `tail`, `shape`, column names, and data types.
- Display only the `'ProductCategory'` and `'Amount'` columns.
- Retrieve the last 3 columns of the DataFrame using `iloc` or `loc`.
- Filter rows where the `'Region'` is 'North' and `'Amount'` is greater than 200.
- Find the value counts for the `'ProductCategory'` column.
- Find the unique values in the `'Region'` column.
- Group by `'Region'` and find the mean amount for each region.

3. Manipulating the DataFrame:

- Modify the `'Amount'` for the transaction with `TransactionID` 102 to 165.
- Add a new column `'Discount'` by calculating 10% of the `'Amount'`.
- Remove the row with `TransactionID` 109.
- Delete the `'Discount'` column from the DataFrame.

Deliverables

Submit a project folder containing:

- Complete the assignment in Google Colab or a Jupyter Notebook.
- For Google Colab, share the notebook link with "View" access. If using Jupyter, upload the notebook to Google Drive and ensure "View" access is enabled.
- Put both the Google Colab notebook in a Google Drive folder and share that folder with view access

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
NumPy Array Creation	Data Structures	Creating 1D & 2D arrays	Beginner
Array Inspection	Data Analysis	Shape, dtype, size	Beginner
Array Operations	Numerical Computing	Conversion, max, min, mean	Beginner
Array Slicing	Data Extraction	Indexing and slicing arrays	Beginner
Pandas Series Creation	Data Structures	Creating labeled data	Beginner
Series Indexing	Data Access	Using index, loc, iloc	Beginner
Series Manipulation	Data Processing	Update, delete, transform values	Beginner
DataFrame Creation	Data Structures	Creating structured datasets	Beginner
Data Exploration	Data Analysis	head(), tail(), shape, info()	Beginner
Data Filtering	Data Handling	Conditional filtering	Beginner
Aggregation	Data Analysis	groupby(), mean(), value_counts()	Intermediate
DataFrame Modification	Data Processing	Add, update, delete columns/rows	Beginner

DA Skill Taxonomy Structure

Skill / Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial Marks Range
NumPy Operations	3	Correct array creation, operations, and slicing	Minor errors in calculations or slicing	1 – 2
Pandas Series	2	Correct creation, indexing, filtering, and manipulation	Minor mistakes in indexing or updates	1
DataFrame Exploration	3	Accurate data exploration and filtering	Partial or incorrect filtering/selection	1 – 2
DataFrame Manipulation	2	Correct updates, additions, and deletions	Minor errors in modifying data	1

DA Skill Taxonomy Structure

- **NumPy Operations:** Array creation, slicing, and numerical computations
- **Pandas Series:** Indexing, filtering, and transformation
- **DataFrame Handling:** Data exploration, filtering, and grouping
- **Data Manipulation:** Updating, adding, and removing data
- **Analytical Thinking:** Extracting insights from datasets

Evaluation Guidelines

Review the submitted **.py / notebook file(.ipynb)** to ensure the following:

- NumPy operations are correctly implemented
- Pandas Series and DataFrame tasks are completed accurately
- Code runs without errors
- Proper structure and readability are maintained

For each section:

- Map work to defined skills
- Assign marks as per the rubric.

Marking Criteria:

- **Full marks:** Correct logic, accurate output, clean code
- **Partial marks:** Minor errors or incomplete implementation

Additional Checks:

- Outputs match expected results
- Code is well-structured and readable
- Proper use of functions and methods

Achievement Criteria & Passing Rules

- Minimum marks required in all sections
- Must demonstrate at least two major skills:
 - **NumPy Operations**
 - **Pandas Data Handling**
- Code must be executable and logically correct
- Provide a **scorecard with feedback** for each submission